Xiyuan Yang

Computer Science Ph.D. Student @ University of Illinois Urbana-Champaign

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Research & Selected Experiences

- My research focuses on agentic foundation models: **AutoResearch**, **World Models** (ICLR 2027, under review), **Optimization** (ICML 2026) and **Trustworthy Agents** (EMNLP 2025, CVPR 2024).
- Have implemented large scale agentic-RL pipelines with **fully async** rollout / training loops, scaling multi-turn, long-horizon **(64k+ tokens traj.)** training across **8+ nodes** on ∼**30B** models.

Education

Ph.D. in Computer Science — Aug 2025 – May 2030 (expected)
University of Illinois Urbana-Champaign — Champaign, USA
Advisor: Prof. Jingrui He

B.Eng. in Computer Science and Technology — Sep 2021 – Jun 2025
Wuhan University — Wuhan, China
Advisor: Prof. Mang Ye

Internship Experience

Google, Google Cloud — Jan 2027 – Apr 2027
Mentor: Shentao Yang — Sunnyvale, USA
Project: Post-Training Long-Horizon Coding Agents

Amazon, Prime Video Group — May 2026 – Aug 2026
Mentor: Zhenyu Liao — Sunnyvale, USA
Project: Scaling Automatic Research Agents via World Models

Microsoft Research, Social Computing Group — Jul 2024 – Feb 2025
Mentor: Fangzhao Wu — Beijing, China
Project 1: Detecting Prompt Injections via Representation Engineering (in EMNLP 2025)
Project 2: Identifying Cross-lingual Inequalities of Foundation Models (in PNAS 2025)
Project 3: Defending Jailbreak Attacks by Vision Language Models (under review of Nature Communications)
Project 4: Measuring Human Contributions in AI-Generated Content (in ACL 2026)
Competition: Foundation Model based News Recommendation (Ranked 1/157 groups)

University of Chicago, Department of Computer Science — Feb 2024 – Sep 2024
Mentor: Prof. Tian Li — Chicago, USA
Project: Differentially Private Distributed Optimization (under review)

Invited Presentations

Rochester Institute of Technology Prof. Haibo Yang's Group (invited talk) — 2026
Automatic Research Agents: Directions and Practice — Rochester, USA

Amazon PV CoREL Journal Club (invited talk) — 2026
Scaling Automatic Research Agents via World Models — Sunnyvale, USA

Jump Trading AI Research Symposium (selected poster, ∼25 worldwide) — 2026
Optimizer-Aware Kernels for Foundation Model Finetuning Dynamics — New York City, USA

Publications

- **Scaling Automatic Research Agents via World Models** [Huggingface Daily Paper #1]
  Xiyuan Yang, Sheikh Sarwar, Jingru Cheng, Zhan Shi, Duanshun Li, Huiyuan Chen, Haiyang Zhang, Chenlei Guo, Jingrui He, Zhenyu Liao
  Under Review, ICLR 2027

- **Code as Agent Harness** [Huggingface Daily Paper #1]
  Xuying Ning et al. (43 authors, incl. **Xiyuan Yang**)
  arXiv 2026

- **Heterogeneous Scientific Foundation Model Collaboration** [Huggingface Daily Paper #1]
  Zihao Li, Jiaru Zou, Feihao Fang, Xuying Ning, Mengting Ai, Tianxin Wei, Sirui Chen, **Xiyuan Yang**, Jingrui He
  arXiv 2026

- **Preconditioning Neural Tangent Kernel for Adaptive Optimization**
  Xiyuan Yang, Wenxuan Bao, Katherine Tieu, Jingrui He
  ICML 2026

- **Latent Collaboration in Multi-Agent Systems** [Huggingface Daily Paper #1]
  Jiaru Zou, Ruizhong Qiu, Gaotang Li, **Xiyuan Yang**, Katherine Tieu, Pan Lu, Ke Shen, Hanghang Tong, Yejin Choi, Jingrui He, James Zou, Mengdi Wang, Ling Yang
  ICML 2026

- **Ramen: Robust Test-Time Adaptation of Vision-Language Models with Active Sample Selection**
  Wenxuan Bao, Yanjun Zhao, **Xiyuan Yang**, Jingrui He
  CVPR 2026

- **Measuring Human Contribution in AI-Assisted Content Generation**
  Yueqi Xie, Tao Qi, Jingwei Yi, **Xiyuan Yang**, Ryan Whalen, Junming Huang, Qian Ding, Yu Xie, Xing Xie, Fangzhao Wu
  ACL 2026

- **Defending LLMs Against Jailbreak Attacks Utilizing Cross-Modality Generalization Gap**
  Xiyuan Yang*, Chenglong Wang*, Haoyu Tang*, Yueqi Xie, Bin Zhu, Lingjuan Lyu, Mang Ye, Fangzhao Wu
  Under Review 2025

- **Uncovering Inequalities in New Knowledge Learning by Large Language Models Across Different Languages**
  Chenglong Wang, Haoyu Tang, **Xiyuan Yang**, Yueqi Xie, Jina Suh, Sunayana Sitaram, Junming Huang, Yu Xie, Pengjun Zhao, Zhaoya Gong, Xing Xie, Fangzhao Wu
  PNAS 2025

- **Defending Against Indirect Prompt Injection by Instruction Detection**
  Tongyu Wen*, Chenglong Wang*, **Xiyuan Yang***, Haoyu Tang, Yueqi Xie, Lingjuan Lyu, Zhicheng Dou, Fangzhao Wu
  EMNLP 2025

- **Differentially Private Federated Clustering with Random Rebalancing**
  Xiyuan Yang, Shengyuan Hu, Soyeon Kim, Tian Li
  Under Review 2025

- **FedAS: Bridging Inconsistency in Personalized Federated Learning**
  Xiyuan Yang, Wenke Huang, Mang Ye
  CVPR 2024
- **Dynamic Personalized Federated Learning with Adaptive Differential Privacy**
  Xiyuan Yang*, Wenke Huang*, Mang Ye
  NeurIPS 2023
- **Robust Heterogeneous Federated Learning under Data Corruption**
  Xiuwen Fang, Mang Ye, **Xiyuan Yang**
  ICCV 2023

Community Contributions

- **Conference Review**: NeurIPS, ICML, ICLR, AAAI, AISTATS, CVPR, ICCV
- **Journal Review**: IEEE TPAMI, IEEE TMC, IEEE TNNLS, IEEE TIFS, IEEE TKDE, IEEE TDSC
- **PFLlib**: Personalized and Distributed Optimization Library — Github Stars **2000+**
- **LatentMAS**: Latent Communication of Multi-agent System — Github Stars **600+**

Awards & Scholarships

- PhD Student Fellowship (first-year), University of Illinois Urbana-Champaign — 2025
- Star of Tomorrow Award, Microsoft Research — 2025
- Overseas Exchange and Study Award, Wuhan University — 2024
- Leijun Computer Science Research Award, Wuhan University — 2024

Skills

- **Machine Learning**: PyTorch, CUDA, HuggingFace Transformers, vLLM, NumPy
- **Systems & Tools**: Linux, Distributed Systems, Docker, Git, LaTeX